3.9.2 Classification of stars (A-level only)

3.9.2.1 Classification by luminosity (A-level only)

Content
Apparent magnitude, m. The Hipparcos scale. Dimmest visible stars have a magnitude of 6. Relation between brightness and apparent magnitude. Difference of 1 on magnitude scale is equal to an intensity ratio of 2.51. Brightness is a subjective scale of measurement.

3.9.2.2 Absolute magnitude, M (A-level only)

Content
Parsec and light year. Definition of M, relation to m: $m - M = 5 \log\frac{d}{10}$

Stefan's law and Wien's displacement law.

General shape of black-body curves, use of Wien's displacement law to estimate black-body temperature of sources.

Experimental verification is not required.

$\lambda_{max}T = constant = 2.9 \times 10^{-3}\ \mathrm{m\ K}$

Assumption that a star is a black body.

Inverse square law, assumptions in its application.

Use of Stefan's law to compare the power output, temperature and size of stars

$P = \sigma A T^4$

3.9.2.4 Principles of the use of stellar spectral classes (A-level only)

Description of the main classes:

Spectral class	Intrinsic colour	Temperature / K	Prominent absorption lines
O	blue	25 000 – 50 000	He^+, He, H
B	blue	11 000 – 25 000	He, H
A	blue-white	7 500 – 11 000	H (strongest) ionized metals
F	white	6 000 – 7 500	ionized metals
G	yellow-white	5 000 – 6 000	ionized & neutral metals
K	orange	3 500 – 5 000	neutral metals
M	red	< 3 500	neutral atoms, TiO

Temperature related to absorption spectra limited to Hydrogen Balmer absorption lines: requirement for atoms in an n = 2 state.

General shape: main sequence, dwarfs and giants.

Axis scales range from –10 to +15 (absolute magnitude) and 50 000 K to 2 500 K (temperature) or OBAFGKM (spectral class).

Students should be familiar with the position of the Sun on the HR diagram.

Stellar evolution: path of a star similar to our Sun on the HR diagram from formation to white dwarf.

3.9.2.6 Supernovae, neutron stars and black holes (A-level only)

Content
Defining properties: rapid increase in absolute magnitude of supernovae; composition and density of neutron stars; escape velocity > c for black holes.
Gamma ray bursts due to the collapse of supergiant stars to form neutron stars or black holes.
Comparison of energy output with total energy output of the Sun.
Use of type 1a supernovae as standard candles to determine distances. Controversy concerning accelerating Universe and dark energy.
Students should be familiar with the light curve of typical type 1a supernovae.
Supermassive black holes at the centre of galaxies.
Calculation of the radius of the event horizon for a black hole, Schwarzschild radius (R_s), $R_s \approx \frac{2GM}{c^2}$